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Astra Blue and Basic Fuchsin Double Staining of Plant Materials

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ABSTRACT. Methods for double staining plant materials using astra blue and basic fuchsin are described here. These methods can be applied to free hand and microtome sections embedded in paraffin, paraplast or historesin. Also, they can be used to study isolated epidermal peels and pollen preparations. Temporary, semipermanent and permanent preparations were studied. Astra blue stained polysaccharides of the cell wall such as cellulose and pectins. Basic fuchsin showed an affinity for lignified, suberized or cutinized walls. The easy preparation of the reagents, excellent color contrast of the histological preparations, and brief staining times of some methods makes them useful for both routine research and didactic purposes. Also, excellent color or black and white photomicrography can be obtained after the double staining described here.

Key words: astra blue, basic fuchsin, double staining, plant material

Phthalocyanine derivatives and compounds are widely used to stain mucopolysaccharides in animal tissues, but they are seldom used to study plant materials (Gahan 1984, Achar et al. 1993). Astra

blue belongs to this group of staining substances and its use in plant histology was proposed by Maác and Vágás (1961), in association with safranin and auramine, to differentiate cellulosic from lignified cell walls. Usually, astra blue was used in association with safranin (Woessner 1970, Lindauer 1971, Bukatsch 1972, Kropp 1972, Gerlach 1984). More recently, Achar et al. (1993), reported the use of some phthalocyanine stains associated with safranin for plant material; astra blue was not described among the assayed stains. Roeser (1972) used astra blue associated with basic fuchsin to stain plant material, although he did not define the stain concentration or explain how it was prepared. Roeser's double staining methodology was used for studying paraffin sections of *Pinus* leaves.

The aim of the current study is to present the use of astra blue and basic fuchsin for free hand or microtome sections of leaf, root, stem and flower, plus isolated epidermal peels and pollen preparations. These double staining methods were tested on different plant materials subjected to different fixation, dehydration, infiltration, and embedding or mounting media.

MATERIALS AND METHODS

Astra blue No. 101278, E. Merck, Germany) and basic fuchsin (C.I. 42510) stains were used throughout this study. These stains were prepared as appropriate for the staining methods employed. Astra blue can be aqueous or acidified with a 2%

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Table 1. Procedures Used for Preparing Samples

Organ	Species	Fixative	Dehydration Series	Embedding
Leaf	<i>Baccharis concinna</i> G. M. Barroso	BNF	NBA	Paraplast
	<i>Caryocar brasiliense</i> Camb.	FAA	TBA	Paraplast
	<i>Eugenia uniflora</i> L.	FAA	Ethanol	Paraffin
	<i>Ficus microcarpa</i> L. f.	FAA	TBA	Paraffin
	<i>Hemerocallis flava</i> L.	70% ethanol	Ethanol	Histoiresin
	<i>Lavoisiera confertiflora</i> Naud.	FAA		
	<i>Lavoisiera glandulifera</i> Naud.	FAA		
	<i>Lavoisiera subulata</i> Triana	FAA		
	<i>Nerium oleander</i> L.	FAA	Ethanol	Paraplast
	<i>Baccharis dracunculifolia</i> D. C.	BNF	Ethanol	Histoiresin
Stem	<i>Lavoisiera cataphrata</i> D. C.	FAA		
	<i>Caryocar brasiliense</i> Camb.	FAA	TBA	Paraplast
Root	<i>Caesalpinia echinata</i> Lam.	70% ethanol		
	<i>Caryocar brasiliense</i> Cam.	FGAA	Ethanol	Histoiresin
	<i>Epidendrum fulgens</i> Brongn.	70% ethanol or FAA		
	<i>Tillandsia recurvata</i> L.	FAA	NBA	Paraffin
Flower	<i>Aylthonia blackii</i> (L. B. Smith) Menezes	FAA	Ethanol	Histoiresin
	<i>Baccharis concinna</i> G. M. Barroso	BNF	NBA	Paraplast
	<i>Barbacenia flava</i> Mart. ex Schult. F. Menezes	Allen-Bouin	TBA	Paraplast
	<i>Hemerocallis flava</i> L.	70% ethanol	Ethanol	Histoiresin
	<i>Vellozia caruncularis</i> Mart. Ex Seub. Menezes	Allen-Bouin	TBA	Paraffin
Pollen	<i>Caryocar brasiliense</i> Camb.	70% ethanol		
	<i>Hemerocallis flava</i> L.	70% ethanol		

Fixatives: Allen-Bouin (Sass 1951); FAA (formalin-acetic acid-50% alcohol; Johansen 1940); FGAA (formalin-glutaraldehyde-acetic acid-95% alcohol; Lersten and Curtis 1988); BNF (buffered neutral formalin; Lillie cited in Clark 1981); 70% ethanol.

Dehydration series: ethanol (Johansen 1940); NBA (n-butyl alcohol; Lang 1937); TBA (tert-butyl alcohol; Johansen 1940).

Embedding: paraffin wax (M. P. 54–56 C; Reagen, Brazil) with 8% beeswax (w/w); Paraplast (Oxford Labware, St. Louis, Mo.); Histoiresin (Leica, Germany).

tartaric acid (Luque et al. 1996); basic fuchsin can be aqueous or in 50% ethanol.

For pollen preparations, glycerol gelatin was used according to Kaiser (1880) in which 1 ml of 0.5% acidified astra blue and 0.5 ml of 1% basic fuchsin were added to 6 ml of glycerol gelatin warmed to 35 C. The different plant samples were processed as described in Table 1.

The histological sections were obtained either by free hand using razor blades or by a rotary microtome. In the latter procedure, the sections were 5–15 mm thick depending on the plant material and embedding media used. The paraffin embedded sections were affixed to slides using

Haupt's (1930) or Bissing's (1974) adhesives. The temporary slides were mounted with 50% glycerin, the semipermanent ones with glycerol gelatin (Kaiser 1880), and the permanent slides with synthetic Canada balsam (Caedax, Merck) diluted in n-butyl acetate (1:1, v/v).

Images of the sections, isolated epidermal peels and pollen preparations were recorded using an optical photomicroscope with a blue filter (LKB) and Kodacolor ASA 100 film.

Hand Sections

Method I, for temporary or semipermanent slides. a) Clear the sections with 10–50%

Figs. 1–6. Staining methods for plant tissues with astra blue and basic fuchsin. Abbreviations: LS, longitudinal section; PS, paradermal section; TS, transverse section.

Fig. 1. Hand section, method I: root of *Caesalpinia echinata*; detail of the periderm and vascular system. TS. $\times 145$.

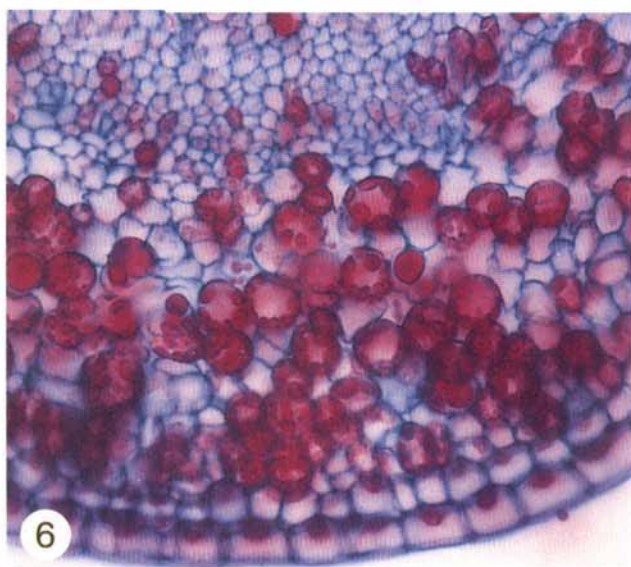
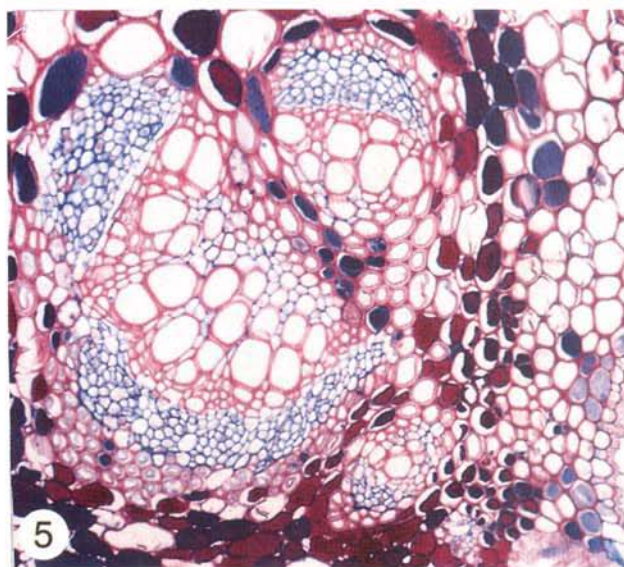
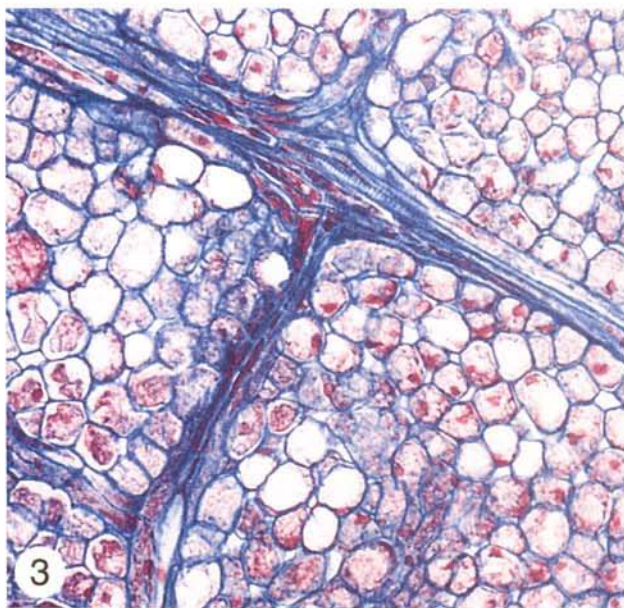
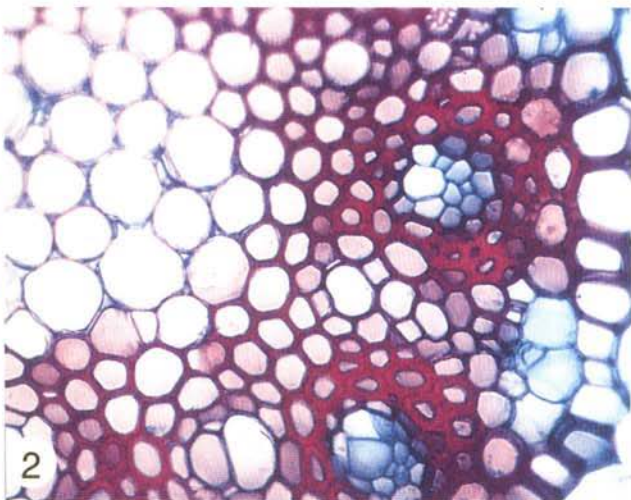
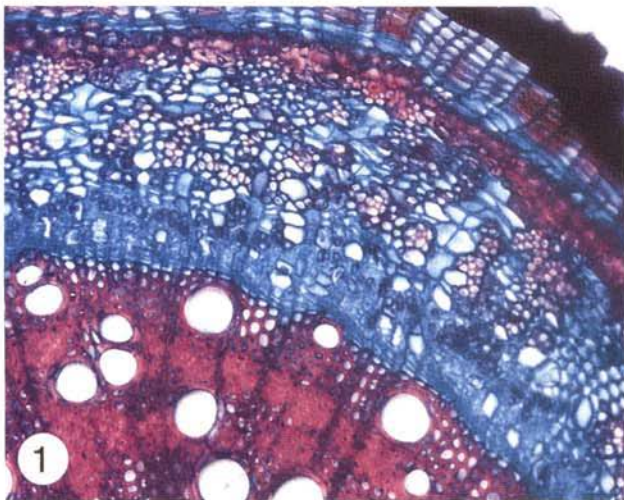
Fig. 2. Hand section, method II: root of *Epidendrum fulgens*; detail of the vascular system. TS. $\times 290$.

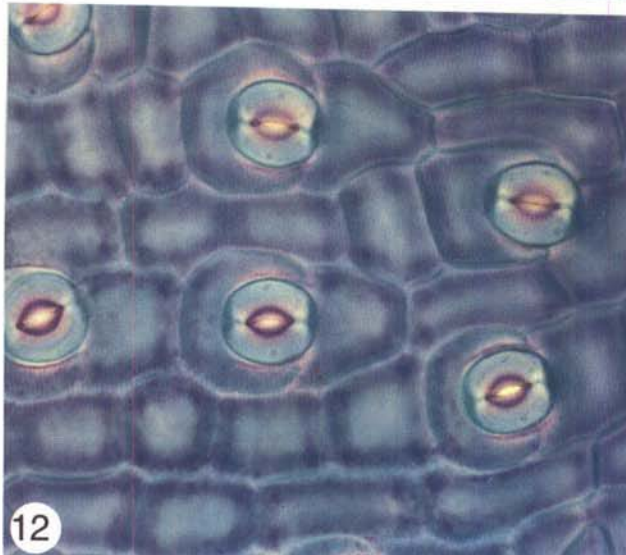
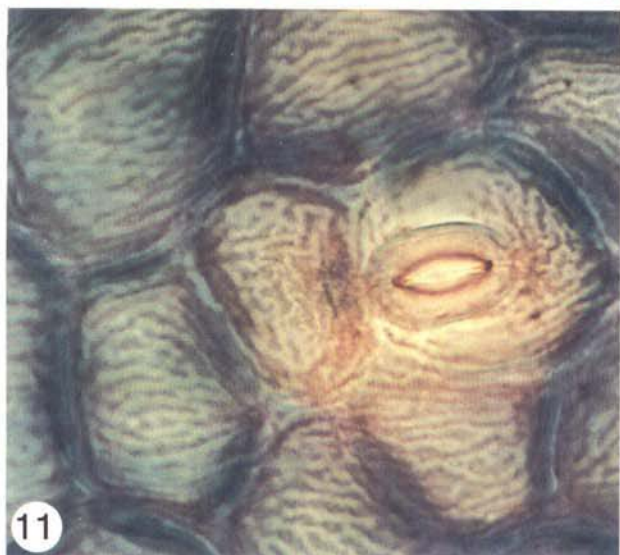
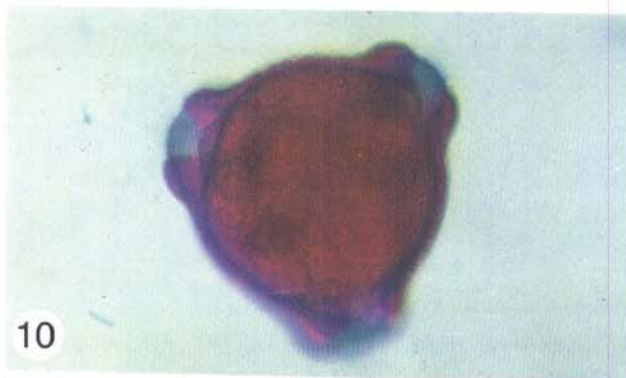
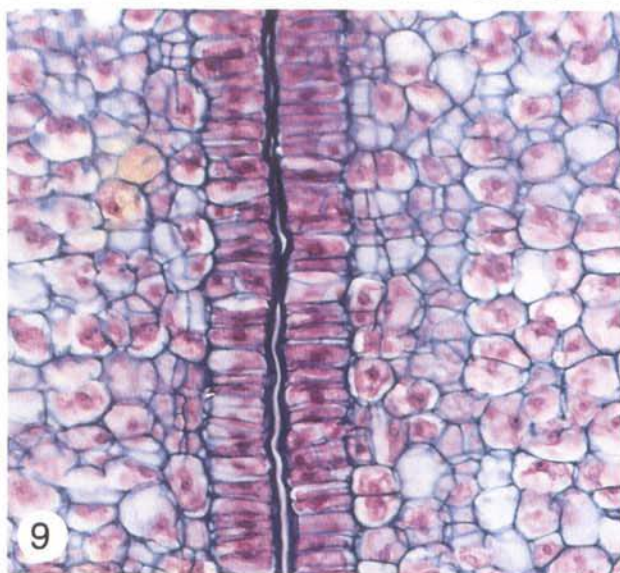
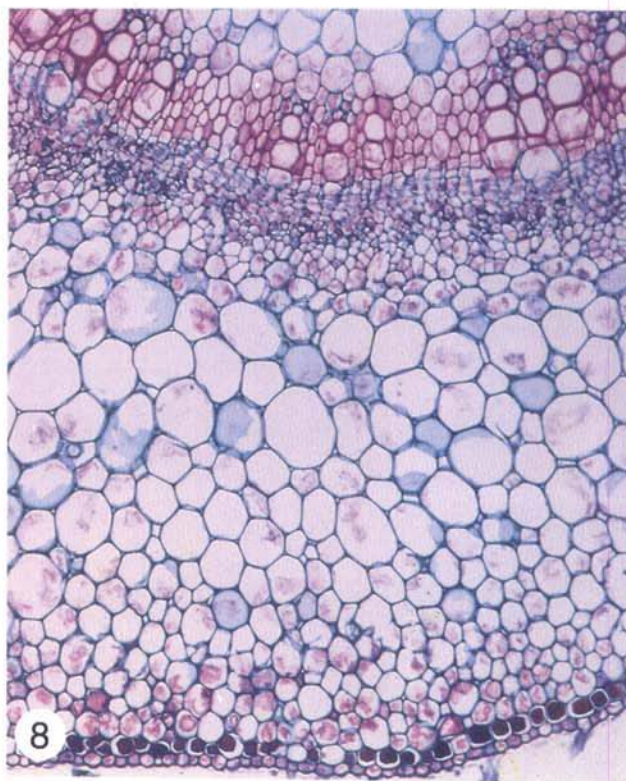
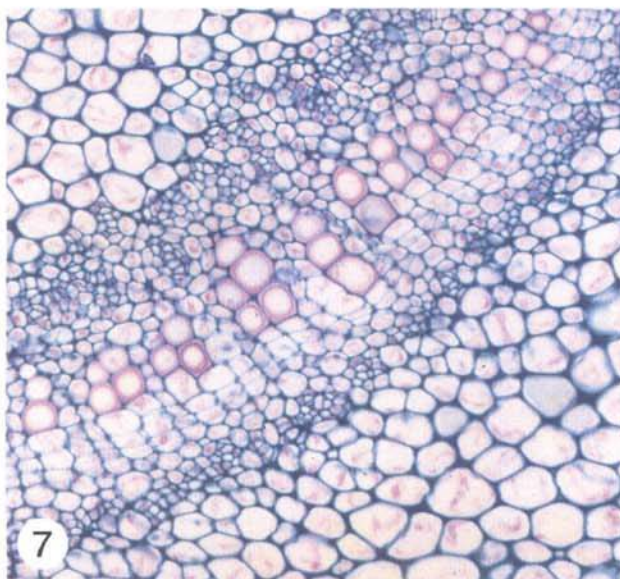
Fig. 3. Paraffin section, method I: leaf of *Eugenia uniflora*; detail of the palisade parenchyma cells and minor veins. PS. $\times 145$.

Fig. 4. Paraffin section, method I: anther of *Vellozia carruncularis*; detail of the microspore mother cells. LS. $\times 580$.

Fig. 5. Paraffin section, method II: leaf of *Caryocar brasiliense*; detail of the midrib. TS. $\times 145$.

Fig. 6. Paraffin section, method II: young leaf of *Ficus microcarpa*; detail of the midrib, parenchyma cells and epidermis. TS. $\times 290$.





commercial household bleach for 2–5 min. Use a higher concentration for hard materials; the time required is determined by observation under a stereomicroscope. b) Wash 3–5 times in distilled water until the preparation becomes odorless. c) Pass through 1% acetic acid. d) Stain in 0.5% acidified astra blue (3–5 min). e) Wash in distilled water. f) Stain in 0.25% ethanolic basic fuchsin for a few seconds. The staining time can be increased when the stain solution is less concentrated. Good results can also be obtained with 0.1% ethanolic basic fuchsin (3–5 min). g) Wash in distilled water. h) Differentiate using saturated picric acid (fast baths for seconds to 1 min). The passage through a saturated picric acid solution can be skipped when the desired contrast between the two stains is reached soon after preparation. Keep dry picric acid in a cool place and away from fire; it may explode when rapidly heated or jarred. Picric acid may cause local or generalized allergic reactions; work in a fume hood. i) Pass through 2–3 distilled water baths until the yellowish color from the picric acid disappears. j) Mount in 50% glycerin or glycerin gelatin.

Method II, for permanent slides. a) Follow the steps described for Method I until item f. b) Dehydrate progressively in an ethanol series (50, 70, 90, and 100%) followed by another passage through 100% ethanol. c) Pass through 100% ethanol:n-butyl acetate (1:1, v/v). n-Butyl acetate is flammable: flash point 22 C, boiling point 125–126 C. It is irritating, may cause conjunctivitis and it is a narcotic in high concentrations; work in a fume hood. n-Butyl acetate can be replaced by xylene substitute (Electron Microscopy Sciences, Fort Washington, Pa). d) Pass through n-butyl

acetate 2 times, 5 min each. e) Mount in Canada balsam diluted in n-butyl acetate.

Paraffin Sections

Method I, for permanent slides. a) Pass through xylene 2 times, 15 min each, until the paraffin or paraplast is totally removed. Xylene is flammable: flash point 29 C, boiling point 125–126 C; it is irritating, carcinogenic and a narcotic in high concentrations; work in a fume hood. Xylene can be replaced by a xylene substitute (Electron Microscopy Sciences). b) Oven dry at 40–45 C for 15–30 min. c) Stain in 1% acidified astra blue for minutes to overnight. d) Wash in distilled water. e) Stain in 0.1% ethanolic basic fuchsin for seconds to minutes; f) Wash in distilled water. g) Differentiate using saturated picric acid (fast baths for seconds to 1 min). h) Wash in distilled water 2–3 times until the yellowish color of the picric acid is completely removed. i) Oven dry at 40–45 C overnight. j) Pass through n-butyl acetate 2 times, 5 min each. k) Mount in Canada balsam diluted in n-butyl acetate.

Method II, for permanent slides. a) Pass through xylene 2 times, 15 min each, until the paraffin or paraplast is completely removed. b) Pass through xylene: 100% ethanol (1:1, v/v). c) Pass through an ethanol series (100, 90, 70 and 50%). d) Wash in distilled water. e) Stain in 1% acidified astra blue for minutes to overnight. Depending on the material, the concentration of acidified astra blue may be lower (0.3%). f) Wash in distilled water. g) Stain in 0.1% ethanolic basic fuchsin for seconds to minutes. A 0.25% aqueous solution of basic fuchsin can be substituted for the ethanolic solution. h) Wash in distilled water. i) Dehydrate progressively in an ethanol

Figs. 7–12. Staining methods for plant tissues with astra blue and basic fuchsin. Abbreviations: LS, longitudinal section; TS, transverse section.

Fig. 7. Paraffin section, method II: leaf of *Nerium oleander*; detail of the midrib. TS. $\times 290$.

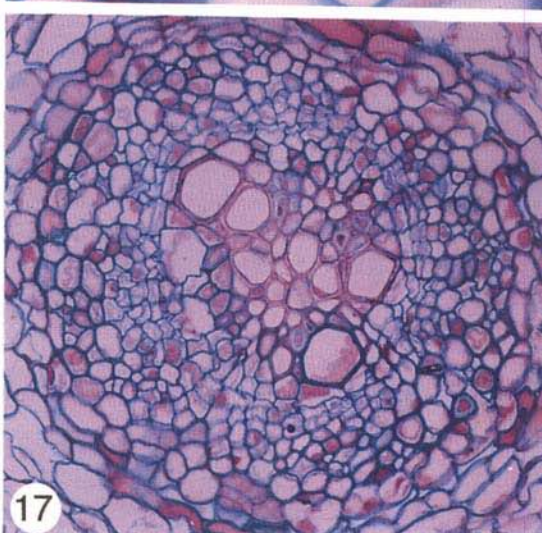
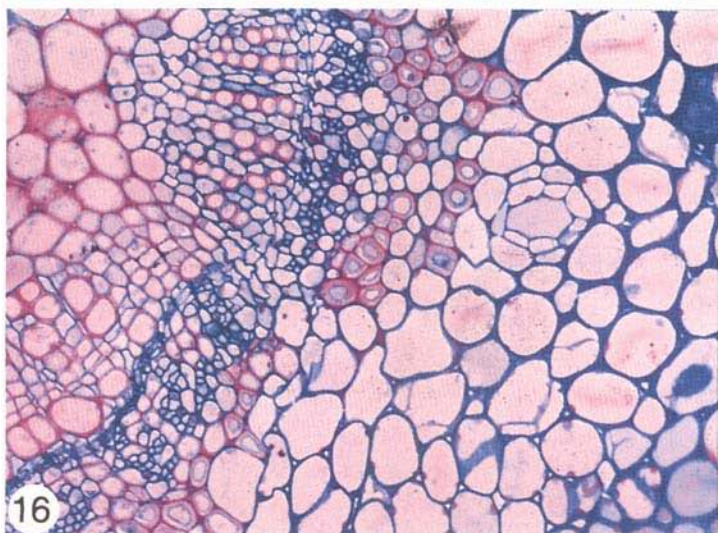
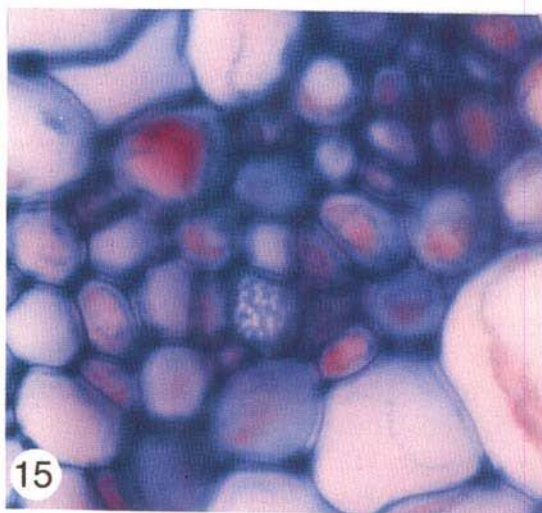
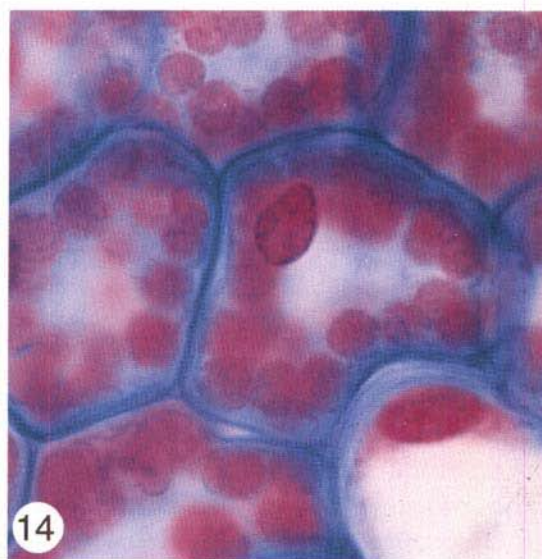
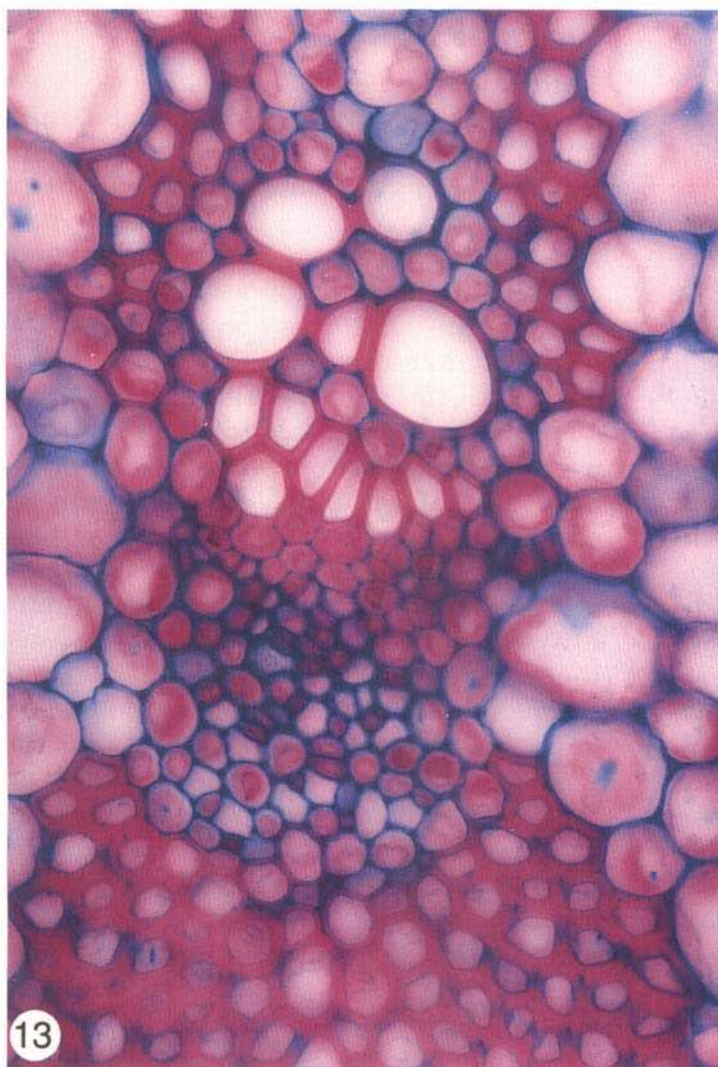
Fig. 8. Paraffin section, method II: stem of *Caryocar brasiliense*; detail of the epidermis, cortex and vascular system. TS. $\times 145$.

Fig. 9. Paraffin section, method II: ovary of *Barbacenia flava*; detail of the secretory tissue of the septal nectarium. LS. $\times 290$.

Fig. 10. Pollen preparation: *Caryocar brasiliense*. $\times 1450$.

Fig. 11. Isolated epidermal peel: leaf of *Lavoisiera subulata*. $\times 580$.

Fig. 12. Isolated epidermal peel: leaf of *L. confertiflora*. $\times 580$.



series (50, 70, 90, and 100%) followed by another passage through 100% ethanol. j) Pass through 100% ethanol:n-butyl acetate (1:1, v/v). k) Pass through n-butyl acetate 2 times, 5 min each. l) Mount in Canada balsam diluted in n-butyl acetate.

Historesin Sections

Method I, for permanent slides. a) Stain in 3% astra blue maintained in an oven at 40–45 C for hours to overnight. If the sections wrinkle, they should be spread again and the slides dried on a heating plate at 40–45 C for 15–30 min. b) Wash in distilled water until excess stain is removed. c) Oven dry overnight at 40–45 C. d) Stain in 0.06% ethanolic basic fuchsin for a few seconds. An aqueous solution of basic fuchsin can be substituted for the ethanolic solution, and in this circumstance item (c) can be excluded. e) Oven dry at 40–45 C overnight. f) Pass through ethanol 100%. The passage through 100% ethanol is performed to remove excess stain from the resin. This passage must be rapid to avoid wrinkling the sections. The stain excess can also be removed with 20–30% ethanol. g) Pass through n-butyl acetate. h) Mount with Canada balsam diluted in n-butyl acetate.

Isolated Epidermal Peels

Method I, for temporary or semipermanent slides. a) Place the epidermal fragments obtained from maceration (Jeffrey, cited in Johansen 1940) in a Petri dish containing distilled water and sweep out the internal face with a brush to remove the cellular debris of the mesophyll. b) Wash in distilled water 3–5 times. c) Pass through 1% acetic acid for 1 min. d) Stain in 0.5% acidified astra blue for 3–5 min. e) Wash in distilled water. f) Stain in 0.25% ethanolic basic fuchsin for 10–20 sec. g) Wash in distilled water. h) Differentiate using satu-

rated picric acid (fast baths for seconds to 1 min). i) Wash in distilled water 2–3 times until the yellowish color of the picric acid is removed. If the material remains overstained by fuchsin, the concentration of this stain can be reduced. j) Mount in 50% glycerin or glycerin gelatin.

Pollen Preparation

Method I, for semipermanent slides. a) Remove the pollen grains from the anthers maintained in 70% ethanol. b) Drip the glycerol gelatin with the stain, previously warmed to 30–35 C, on a microscope slide containing the pollen. c) Place the slide on a heating plate at 30–35 C for 5 min. d) Cover with a previously warmed glass coverslip. e) Examine the material after 30–60 min. The color intensity depends on the stain penetration into the pollen grains; thus, a waiting period of 30–60 min is required to obtain satisfactory preparations.

RESULTS AND DISCUSSION

The results of the methods described in this paper are shown in Figs. 1–17.

Double staining is frequently used in plant histology to show cell walls, cytoplasmic contents and other cell components. In the present study we have demonstrated that the combination of astra blue and basic fuchsin produces excellent contrast and that this double staining procedure can be applied to various plant samples. Furthermore, we have shown the association of these stains to be adequate for different techniques used in plant histology regardless of the type of fixation, dehydration, infiltration, embedding, sectioning, and mounting methods used.

According to Beneš and Uhlirová (cited in Gahan 1984) and Beneš (1968), alcian blue and astra blue stain pectins; how-

Figs. 13–17. Staining methods for plant tissues with astra blue and basic fuchsin; transverse sections.

Fig. 13. Historesin section: leaf of *Hemerocallis flava*; detail of the vascular bundle. $\times 580$.

Fig. 14. Historesin section: leaf of *H. flava*; detail of the mesophyll cells showing nucleus and chloroplasts. $\times 1450$.

Fig. 15. Historesin section: leaf of *H. flava*; detail of the phloem tissue. $\times 1450$.

Fig. 16. Historesin section: stem of *Baccharis dracunculifolia*; detail of the cortex and vascular system. $\times 290$.

Fig. 17. Historesin section: root of *Caryocar brasiliense* showing vascular system. $\times 290$.

ever, after pectinase digestion and at different pH, they concluded that those stains did not stain methylated or esterified pectins.

Studies reported by Jeffree (1993) involving immobilization of plant polysaccharides on blotting membranes demonstrated that soluble and acid polysaccharides (pectins, carboxymethylcellulose, etc.) are stained by cationic stains, e.g., alcian blue 8GX, while the neutral polysaccharides such as xyloglucans, are stained by PAS. Schumacher and Adam (1994) confirmed that glycosaminoglycans can be identified by different histochemical procedures including staining by alcian blue and its analogs. Luque et al. (1996), working with xerophytic, mesophytic, and aquatic plants, showed that the staining with astra blue is very similar to that obtained with alcian blue 8GX, leading us to infer that astra blue stained the acid polysaccharides of the cellular wall.

The mechanism by which astra blue binds to tissues is controversial. Maác and Vágás (1961), Beneš and Uhlirová (cited in Gahan 1984) and Beneš (1968) suggested that the staining is probably due to ionic bonds between the stain and the free carboxyl groups of the pectin compounds. Achar et al. (1993), however, proposed that the chromophore of the phthalocyanine complex possibly interacts with the plant tissue via hydrogen bonding, charge transfer through the central metal atom, or a combination of these forces.

Basic fuchsin showed affinity for lignified and for suberized or cutinized walls of the plant tissues. Other structures devoid of lignin, such as the chloroplasts, were also stained by basic fuchsin. Thus, we propose that the basic fuchsin demonstrated the phenolic substances, among them lignin, but that it also has affinity for other structures with different compositions. The results obtained with basic fuchsin are similar to those found with safranin, according to Berlyn and Miksche (1976), which stains lignified or cutinized walls and chromosomes.

The preparations using the association of astra blue and basic fuchsin provided consistent color and excellent contrast between the different cell structures, thus allowing good histological analysis and color or black and white photomicrographs.

It is important to stress that the time for staining varies with different methods, plant materials, and even with different organs or parts of the same material. Thus, it is very important to define the optimal length of time for staining each material and/or to adjust the stain concentrations to obtain the best results. It should be emphasized also that the result obtained with Historesin sections stained with toluidine blue O, the most commonly used stain, is inferior to astra blue and basic fuchsin with regard to structural clarity, especially cell walls.

The easy preparation of the reagents and the quickness of some methods employing our double staining recommend its use for both routine research and didactic purposes.

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